

Campus Event Management System

Technical Architecture & Database Aggregation Report

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Repository:	https://github.com/akhill-reddy/campus-event-manager		
Tech Stack:	Node.js, Express.js, MongoDB Atlas, EJS, Bootstrap 5		

1. EXECUTIVE SUMMARY & SYSTEM ARCHITECTURE

The **Campus Event Manager** is a multi-tier full-stack application built to streamline university event scheduling, participant tracking, and real-time database aggregations using Node.js, Express, and MongoDB Atlas.

Layer	Technology	Implementation Details
Backend Runtime	Node.js / Express.js	Handles REST routing, middleware request validation, and dynamic view rendering.
Database Store	MongoDB Atlas / Mongoose	Document persistence store managing relational references and embedded participant arrays.
Presentation	EJS & Bootstrap 5	Server-side templating providing responsive layouts and component styling.
Analytics Engine	MongoDB Aggregations	Processes multi-stage pipeline operations (\$unwind, \$group, \$lookup, \$project).

2. SYSTEM USER INTERFACES & MODULES

2.1 Main Operations Dashboard



Figure 1: Main Platform Dashboard showing live operational counters and popular event metrics.

2.2 Events Catalog & Management



Figure 2: Events Catalog displaying active workshop cards, categories, and capacity indicators.

2.3 Event Details & Participant Registration



Figure 3: Event Detail metadata view featuring venue details and administrative controls.



Figure 4: Participant registration interface and active attendee roster management table.

2.4 User Directory & Role Management



Figure 5: User Directory table with role badges, department mappings, and history tracking links.

3. MONGODB AGGREGATION PIPELINES & ANALYTICS

3.1 Analytics & Reporting Interface



Figure 6: Analytics Dashboard displaying category metrics, popularity rankings, and inactive user tracking.

Pipeline 1: Total Registrations by Category

```
db.events.aggregate([
  { $unwind: "$registrations" },
  { $group: { _id: "$category", totalEvents: { $addToSet: "$_id" }, confirmedRegistrations: { $sum: 1 } } },
  { $project: { category: "$_id", eventsCount: { $size: "$totalEvents" }, confirmedRegistrations: 1, _id: 0 } },
  { $sort: { confirmedRegistrations: -1 } }
]);
```

Pipeline 2: Top 5 Events by Occupancy Rate

```
db.events.aggregate([
  { $project: { title: 1, category: 1, capacity: 1, registrationsCount: { $size: { $ifNull: ["$registrations", []] } }, occupancyRate: { $multiply: [ { $divide: [ { $size: { $ifNull: ["$registrations", []] } }, "$capacity" ] }, 100 ] } } },
  { $sort: { occupancyRate: -1 } },
  { $limit: 5 }
]);
```

Pipeline 3: Inactive Users Query (\$lookup)

```
db.users.aggregate([
  { $lookup: { from: "events", localField: "_id", foreignField: "registrations.userId", as: "userEvents" } },
  { $match: { userEvents: { $size: 0 } } },
  { $project: { firstName: 1, lastName: 1, email: 1, role: 1 } }
]);
```

4. PROJECT VERIFICATION

Developer	Chamakura Akhil Reddy
Repository Link	https://github.com/akhill-reddy/campus-event-manager
Status	Fully Operational & Verified